

Example: weights of Basset Hounds follows a $N(\mu_1, \sigma_1^2)$ distribution. 04/22 W ①

... labs ... $N(\mu_2, \sigma_2^2)$ distribution.

Assume $\sigma_1 = \sigma_2$.

Data: B: $n_1 = 10$ $\bar{x}_1 = 58.9$ $s_1 = 6.0$

L: $n_2 = 12$ $\bar{x}_2 = 57.4$ $s_2 = 6.5$

a) 90% CI for $\mu_1 - \mu_2$.

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= (58.9 - 57.4) \pm \underset{1.725}{t_{0.05, 20}} \times 6.280 \sqrt{\frac{1}{10} + \frac{1}{12}}$$

$$= (-3.138, 6.138)$$

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(10-1) \times 6^2 + (12-1) \times 6.5^2}{10+12-2}} = \boxed{6.280}$$

Is $\mu_1 = \mu_2$ a reasonable guess based on 90% CI?

Yes. 0 is in 90% CI.

b) Test. $H_0: \mu_1 = \mu_2$.

vs. $H_a: \mu_1 \neq \mu_2$ at $\alpha = 0.1$

$H_0: \mu_1 - \mu_2 = 0$

vs. $H_a: \mu_1 - \mu_2 \neq 0$

1) Test statistic.

$$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(58.9 - 57.4) - 0}{6.280 \sqrt{\frac{1}{10} + \frac{1}{12}}} = 0.558$$

2) Critical value

$$t_{\frac{\alpha}{2}, n_1+n_2-2} = t_{0.05, 20} = 1.725$$

3> Decision and conclusion.

(2)

$$|t^*| = 0.558 < 1.725 = t_{\frac{\alpha}{2}, n_1+n_2-2}.$$

Therefore, fail to reject H_0 .

We don't have strong enough evidence to suggest $\mu_1 \neq \mu_2$.

$$c> \text{p-value} = 2 P(t_{n_1+n_2-2} > |t^*|)$$

$$= 2 P(t_{20} > 0.558)$$

$$\in 2 \cdot (0.20, 0.30)$$

$$\in (0.4, 0.6)$$

$$\text{p-value} \in (0.4, 0.6) > 0.1 = \alpha.$$

Therefore, fail to reject H_0 .

Inference for a population proportion p .

p = population/distribution true proportion.

$\hat{p}$ = sample proportion (point estimate of p)

1. Wald CI.

$$\text{requirements} \begin{cases} 1> n \geq 30 \\ 2> n\hat{p} \geq 5 \\ 3> n(1-\hat{p}) \geq 5 \end{cases}$$

$(1-\alpha) 100\%$ Wald CI. is

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(3)

Notes: 1) $\hat{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

2) Margin of error

$$MOE = Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

MOE only depends on sample size n , not population size.

Example: A R.S 100 kids had 22 with imaginary friends.

$$\hat{p} = \frac{22}{100} = 0.22.$$

Define p = population proportion of all kids with IF. (unknown)

a) Find an approximate 95% Wald CI. for p .

$$\begin{aligned} \hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} &= 0.22 \pm 1.96 \sqrt{\frac{0.22(1-0.22)}{100}} \\ &= (0.139, 0.301) \end{aligned}$$